

Experiment 8

Objective:

To generate the waveform of Phase Shift Keying (PSK) and Frequency Shift Keying signal using MATLAB

Theory:

1. Generation of PSK

Generation of PSK signal PSK is a digital modulation scheme that conveys data by changing, or modulating, the phase of a reference signal (the carrier wave). PSK uses a finite number of phases, each assigned a unique pattern of binary digits. Usually, each phase encodes an equal number of bits. Each pattern of bits forms the symbol that is represented by the particular phase. The demodulator, which is designed specifically for the symbol-set used by the modulator, determines the phase of the received signal and maps it back to the symbol it represents, thus recovering the original data. In a coherent binary PSK system, the pair of signals $S_1(t)$ and $S_2(t)$ used to represent binary symbols 1 & 0 are defined by

$$S_1(t) = \sqrt{2E_b/T_b} \cos 2\pi f_c t$$

$$S_2(t) = \sqrt{2E_b/T_b} (2\pi f_c t + \pi) = -\sqrt{2E_b/T_b} \cos 2\pi f_c t \quad \text{where } 0 \leq t < T_b \text{ and}$$

E_b = Transmitted signed energy for bit

The carrier frequency $f_c = n/T_b$ for some fixed integer n .

Antipodal Signal:

The pair of sinusoidal waves that differ only in a relative phase shift of 180° are called antipodal signals.

BPSK Transmitter

The input binary symbols are represented in polar form with symbols 1 & 0 represented by constant amplitude levels $\sqrt{E_b}$ & $-\sqrt{E_b}$. This binary wave is multiplied by a sinusoidal carrier in a product modulator. The result is a BPSK signal as shown in figure 8.1.

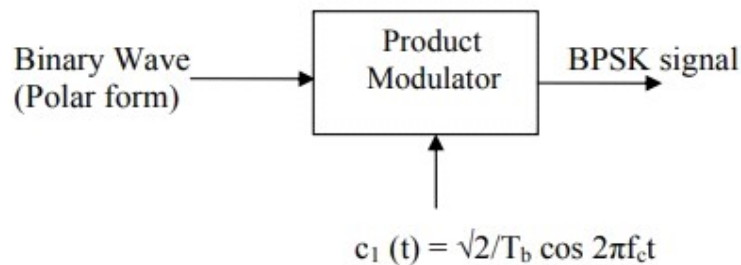


Fig. 8.1 Block Diagram of BPSK Transmitter

2. Generation of FSK

Frequency-shift keying (FSK) is a frequency modulation scheme in which digital information is transmitted through discrete frequency changes of a carrier wave. The simplest FSK is binary FSK (BFSK). BFSK uses a pair of discrete frequencies to transmit binary (0s and 1s) information. With this scheme, the "1" is called the mark frequency and the "0" is called the space frequency. In binary FSK system, symbol 1 & 0 are distinguished from each other by transmitting one of the two sinusoidal waves that differ in frequency by a fixed amount.

$$S_i(t) = \sqrt{2E_b/T_b} \cos 2\pi f_i t$$

$$0 \leq t \leq T_b$$

$$0 \leq t \leq T_b$$

$$\text{elsewhere}$$

where $i=1, 2$ & E_b =Transmitted energy/bit

Transmitted freq= $f_i = (n_c + i)/T_b$, and n = constant (integer), T_b = bit interval

Symbol 1 is represented by $S_1(t)$ and Symbol 0 is represented by $S_0(t)$

BFSK Transmitter

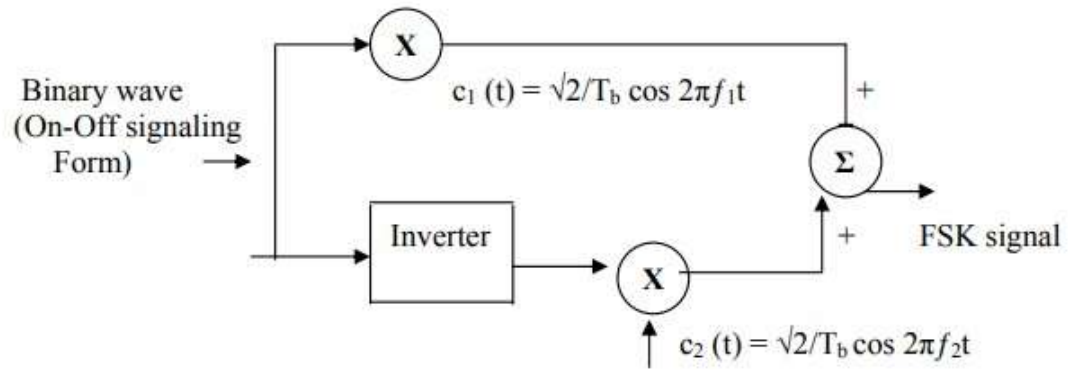


Fig. 8.2 Block Diagram of BFSK Transmitter

The input binary sequence is represented in its ON-OFF form, with symbol 1 represented by constant amplitude of $\sqrt{E_b}$ with & symbol 0 represented by zero volts. By using inverter in the lower channel, we in effect make sure that when symbol 1 is at the input, The two frequency f_1 & f_2 are chosen to be equal integer multiples of the bit rate $1/T_b$. By summing the upper & lower channel outputs, we get BFSK signal as shown in figure 8.2.

PSK modulation:

Program:

```
% PSK modulation
clc;
clear all;
close all;

% Generate Carrier Signal
Tb = 1;
t = 0:Tb/100:Tb;
fc = 2;
c = sqrt(2/Tb) * sin(2*pi*fc*t);

% Generate message signal
N = 8; % Number of bits
m = rand(1, N); % Random binary data
t1 = 0;
t2 = Tb;

for i = 1:N
    t = [t1:0.01:t2]; % Time interval for each bit
    if m(i) > 0.5
        m(i) = 1;
        m_s = ones(1, length(t)); % Message signal (1)
    else
        m(i) = 0;
        m_s = -1 * ones(1, length(t)); % Message signal (-1)
    end

    message(i,:) = m_s;

    % Product of carrier and message signal (BPSK modulation)
    sig(i,:) = c .* m_s;

    % Plot the message signal (Polar form)
    subplot(5, 1, 2);
    axis([0 N -2 2]);
    plot(t, message(i,:), 'r');
    title('Message signal (Polar form)');
    xlabel('t ---->');
    ylabel('m(t)');
```

```

grid on;
hold on;

% Plot the BPSK modulated signal
subplot(5, 1, 4);
plot(t, sig(i,:)); % Change from 'bpsk_sig' to 'sig'
title('BPSK signal');
xlabel('t ---->');
ylabel('s(t)');
grid on;
hold on;

% Update time interval for the next bit
t1 = t1 + 1.01;
t2 = t2 + 1.01;
end

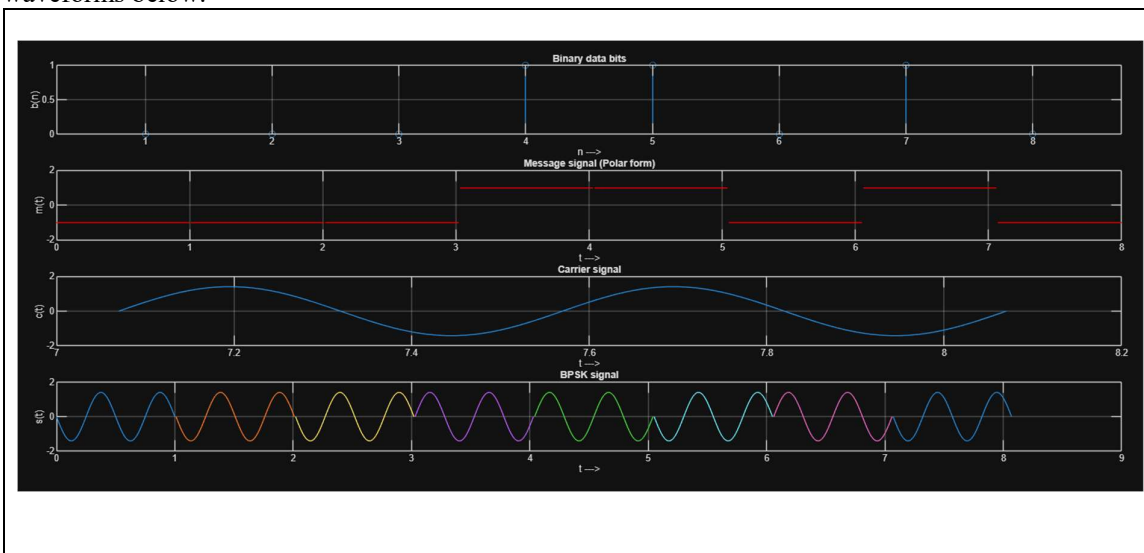
hold off;

% Plot the input binary data
subplot(5, 1, 1);
stem(m);
title('Binary data bits');
xlabel('n ---->');
ylabel('b(n)');
grid on;

% Plot the carrier signal
subplot(5, 1, 3);
plot(t, c);
title('Carrier signal');
xlabel('t ---->');
ylabel('c(t)');
grid on;

```

Attach the desired BPSK waveforms i.e. binary data, message signal, carrier signal and output waveforms below.



% FSK Modulation

```

clc;
clear all;
close all;

```

%Generate Carrier Signal

```
Tb=1;
fc1=2;fc2=5; t=0:(Tb/100):Tb;
c1=sqrt(2/Tb)*sin(2*pi*fc1*t);
c2=sqrt(2/Tb)*sin(2*pi*fc2*t);
```

%generate message signal

```
N=8;
m=rand(1,N);
t1=0;t2=Tb for
i=1:N
t=[t1:(Tb/100):t2]
if m(i)>0.5
m(i)=1;
m_s=ones(1,length(t));
inv_m_s=zeros(1,length(t));
else m(i)=0;
m_s=zeros(1,length(t));
inv_m_s=ones(1,length(t));
end message(i,:)=m_s;
```

%Multiplier

```
fsk_sig1(i,:)=c1.*m_s;
fsk_sig2(i,:)=c2.*inv_m_s;
fsk=fsk_sig1+fsk_sig2;
```

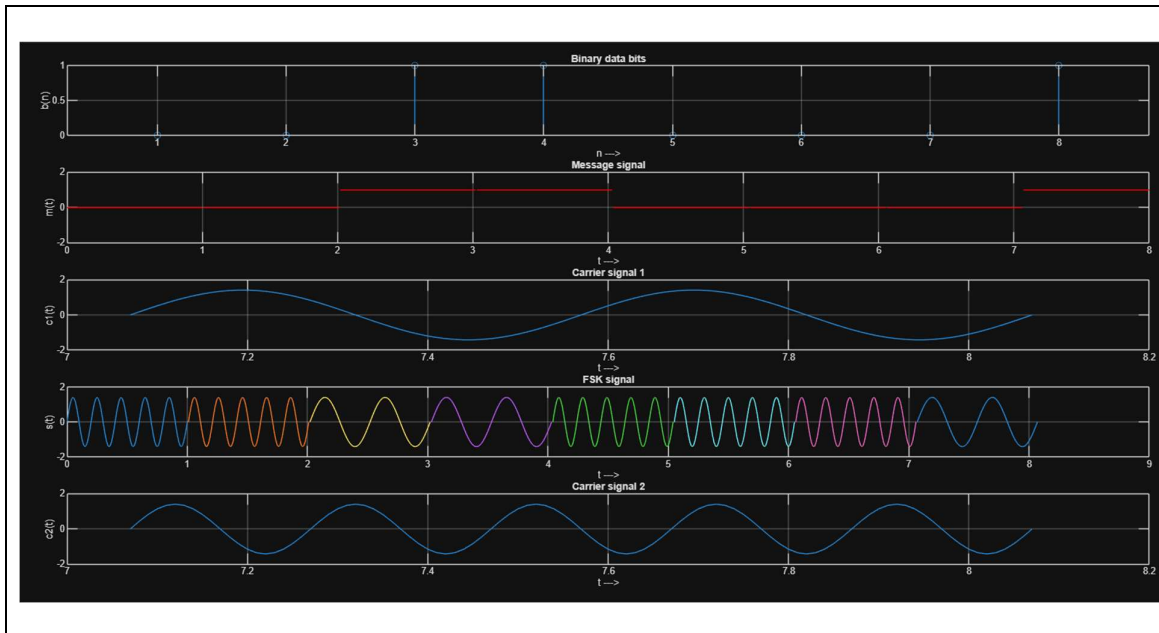
%plotting the message signal and the modulated signal

```
subplot(3,2,2);
axis([0 N -2 2]);
plot(t,message(i,:), 'r');
title('message signal');
xlabel('t---->');
ylabel('m(t)');
grid on; hold on;
subplot(3,2,5);
plot(t,fsk(i,:));
title('FSK signal');
xlabel('t---->');
ylabel('s(t)');
grid on;
hold on;t1=t1+(Tb+.01); t2=t2+(Tb+.01); end
hold off
```

%Plotting binary data bits and carrier signal

```
subplot(3,2,1);
stem(m);
title('binary data');
xlabel('n---->');
ylabel('b(n)');
grid on;
subplot(3,2,3);
plot(t,c1);
title('carrier signal-1');
xlabel('t---->');
ylabel('c1(t)');
grid on;
subplot(3,2,4);
plot(t,c2);
title('carrier signal-2');
xlabel('t---->');
ylabel('c2(t)');
grid on;
```

Attach the desired BFSK waveforms i.e. binary data, message signal, carrier signal 1&2 and output waveforms.



Question:

1. Write down the drawback of DPSK compared to BPSK?

The main drawback is that DPSK has about 3 dB worse bit error rate performance than BPSK, since each bit decision depends on both the current and previous symbols, making it more sensitive to noise and causing potential error propagation.

2. Why do we make 180 degree phase shift in PSK and why not 90 or 270? Comment on this.

As we are representing two symbols (binary 0 and 1), in PSK we use a 180-degree shift to be as far apart as possible on the phase circle to maximize noise tolerance and minimize bit errors.

A 180-degree separation means the two signal states are on exactly opposite sides of the constellation diagram, giving the maximum Euclidean distance between them. This makes them easiest to distinguish even in noisy conditions.

If we used 90-degrees or 270-degrees, the phase states would be closer together, reducing that distance and making the receiver more likely to confuse one symbol for the other, increasing the error rate.

3. Compare FSK, PSK and ASK.

In Amplitude Shift Keying (ASK), digital data is represented by varying the amplitude of a carrier signal while keeping its frequency and phase constant. It is the simplest form of digital modulation and easy to implement, but it is also the most vulnerable to noise because amplitude is easily affected by channel disturbances like fading and interference. As a result, ASK performs poorly in noisy environments but is still used in low-cost and short-range communication systems such as optical fiber links and RFID.

In Frequency Shift Keying (FSK), the frequency of the carrier signal is changed to represent binary data, typically one frequency for bit 1 and another for bit 0. FSK is more resistant to noise than ASK since frequency is less likely to be distorted by amplitude variations. However, it requires a larger bandwidth and slightly more complex transmitter and receiver designs. It is

commonly used in radio transmission, modems, and telemetry systems where reliability is more important than bandwidth efficiency.

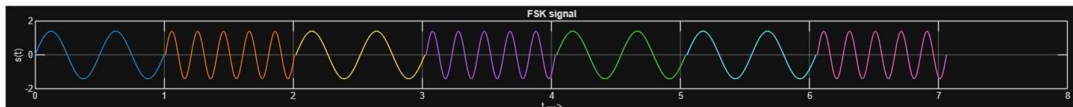
In Phase Shift Keying (PSK), the phase of the carrier signal is varied to represent digital data while keeping amplitude and frequency constant. It provides the best noise immunity and power efficiency among the three, as phase information is relatively unaffected by amplitude noise. However, PSK requires precise carrier synchronization between transmitter and receiver, making it more complex to implement. Because of its high efficiency and robustness, PSK (and its variants like BPSK and QPSK) is widely used in modern wireless, satellite, and high-speed communication systems.

4. Determine the bandwidth and baud for BFSK signal with mark frequency of 49 kHz, a space frequency of 51 kHz, and a bit rate of 2 Kbps.

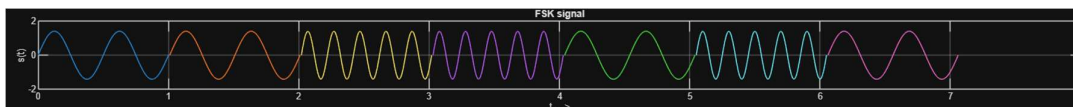
- For BFSK, the baud rate (R_{baud}) is equal to the bit rate (R_b), as each symbol carries one bit of information, that is, $R_{\text{baud}} = R_b$, thus **$R_{\text{baud}} = 2,000 \text{ baud}$** .
- We use Carson's Rule to determine the minimum bandwidth (B):
Carson's Rule: $B = \Delta f + 2R_b$, where Δf is the frequency separation
 $\Delta f = |f_s - f_m|$, where f_s is space frequency and f_m is mark frequency
Thus,
 $\Delta f = |51 \text{ kHz} - 49 \text{ kHz}| = 2 \text{ kHz}$
Finally,
 $B = 2 \text{ kHz} + (2 \times 2000)$
 $B = 6 \text{ kHz}$

5. Sketch the FSK waveform for the input a) 1010110 b) 1100101.

For 1010110:



For 1100101:



Name & ID: Syed Ali Nisar Shah (ss07136) & Syed Muhammad Asad Ali (sa09475)

Date: October 31, 2025

EE-322L Analog and Digital Communications Lab

Fall 2025

Habib University

Dhanani School of Science & Engineering



Lab#08 Phase Shift Keying (PSK), Amplitude and Frequency Shift Keying

Lab #08 Marks distribution:

		LR1=20	LR3=10	LR5=30	AR4=20
In-Lab Tasks	Task 1	/10	/5	/10	/20
	Task 2	/10	/5	/10	
	Task 3			/10	
Marks Obt.	/80				

Lab Evaluation Assessment Rubric

#	Assessment Elements	(0<Level 1<=4)	(4< Level 2<=6)	(6< Level 3<=8)	(8< Level 4<=10)
LR1	Neat and Clean Circuit layout	Components are wired and didn't show neat and clean connections and minimal efforts shown.	Components are wired with untidy connection and didn't show neat and clean connections	Few but not all Components are wired with neat and clean connections	Complete components are wired with neat and clean tight connections. Given task completed in due time.
LR3	Troubleshooting	Unable to identify the fault/minimal effort shown	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and able to make necessary steps and actions to correct it.
LR5	Results & Plots	Figures, graphs, tables contain errors or are poorly constructed, have missing titles, captions, units missing or incorrect, etc.	Most figures, graphs, tables OK, some still missing some important or required features	All figures, graphs, tables are correctly drawn, but some have minor problems or could still be improved	All figures, graphs, tables are correctly drawn and contain titles/captions.
AR4	*Report Submission	Late submission after 1 week.	Late submission after 4 days and within a week.	Late submission after the lab timing and after 2 days of the due date.	Timely submission of the report and in lab time.

***Report:** Report will not be accepted after due date